

prepare_tcga

May 11, 2026

```
[67]: import pandas as pd
import os
import shutil
from pathlib import Path
import glob
from tqdm import tqdm
import numpy as np
```

0.0.1 Clinical data available in the TCGA

The grading of each tumor as well as its diagnostic are based on the WHO 2015 classification.

```
[4]: TCGA_GBM_2015 = pd.read_csv("Clinical-Data_TCGA-GBM_WHO-2015.csv", sep=',').
    ↪drop(columns=["Unnamed: 0"])
TCGA_LGG_2015 = pd.read_csv("Clinical-Data_TCGA-LGG_WHO-2015.csv", sep=',').
    ↪drop(columns=["Unnamed: 0"])

print("Number of samples for the LGG (WHO 2015): ", len(TCGA_LGG_2015), "␣
    ↪patients")
print("Number of samples for the GBM (WHO 2015): ", len(TCGA_GBM_2015), "␣
    ↪patients")
print("Total number of samples: ", len(TCGA_LGG_2015)+len(TCGA_GBM_2015), "␣
    ↪patients")
```

Number of samples for the LGG (WHO 2015): 516 patients

Number of samples for the GBM (WHO 2015): 606 patients

Total number of samples: 1122 patients

0.0.2 Patients with radiomics data as identified by Bakas, et al. 2017

The classification in low- and high-grade glioma is likely based on the same WHO 2015 criteria. In Bakas, et al (2017) there is no mention whatsoever to the genetic profiling of the tumors that they identified. The identification of this subset of patients is solely based on available radiomics data.

```
[5]: TCGA_GBM_2015_radiomics = pd.read_csv("TCGA-GBM_WHO-2015_with-Radiomics.csv",␣
    ↪sep=',')
TCGA_LGG_2015_radiomics = pd.read_csv("TCGA-LGG_WHO-2015_with-Radiomics.csv",␣
    ↪sep=',')
```

```

print("Number of samples with radiomics for the LGG (WHO 2015): ",
      len(TCGA_LGG_2015_radiomics), " patients")
print("Number of samples with radiomics for the GBM (WHO 2015): ",
      len(TCGA_GBM_2015_radiomics), " patients")
print("Total number of samples: ",
      len(TCGA_LGG_2015_radiomics)+len(TCGA_GBM_2015_radiomics), " patients")

```

Number of samples with radiomics for the LGG (WHO 2015): 199 patients

Number of samples with radiomics for the GBM (WHO 2015): 262 patients

Total number of samples: 461 patients

Let's localize the subjects of this subset withn the full TCGA descriptors.

```

[6]: ok = True
problems_GBM = []
for sub_radiomics in TCGA_GBM_2015_radiomics["TCGA ID"].values:
    total = 0
    for sub_total in TCGA_GBM_2015["patient"].values:
        if sub_radiomics == sub_total:
            total += 1
    if total != 1:
        ok = False
        print(f"{sub_radiomics} appears {total} times in the full data_
        ↳description of the GBM group (WHO 2015)")
        problems_GBM.append(sub_radiomics)
if ok:
    print("The subset of GBM patients with radiomics data is uniquely contained_
    ↳in the GBM group of the full clinical sample.")

print("\n")
ok = True
problems_LGG = []
for sub_radiomics in TCGA_LGG_2015_radiomics["TCGA ID"].values:
    total = 0
    for sub_total in TCGA_LGG_2015["patient"].values:
        if sub_radiomics == sub_total:
            total += 1
    if total != 1:
        ok = False
        print(f"{sub_radiomics} appears {total} times in the full data_
        ↳description of the LGG group (WHO 2015)")
        problems_LGG.append(sub_radiomics)
if ok:
    print("The subset of LGG patients with radiomics data is uniquely contained_
    ↳in the LGG group of the full clinical sample.")

```

The subset of GBM patients with radiomics data is uniquely contained in the GBM

group of the full clinical sample.

TCGA-EZ-7264A appears 0 times in the full data description of the LGG group (WHO 2015)

TCGA-EZ-7265A appears 0 times in the full data description of the LGG group (WHO 2015)

And now let's localize the subset with identified pre-operative scans.

```
[7]: TCGA_GBM_2015_radiomics_imaging = TCGA_GBM_2015_radiomics.  
      ↪loc[TCGA_GBM_2015_radiomics["We Identified Baseline Pre-operative_  
      ↪scans"]=="Yes"]  
TCGA_GBM_2015_radiomics.groupby(["We Identified Baseline Pre-operative scans"]).  
      ↪count()
```

```
[7]:
```

	Collection	TCGA ID	\
We Identified Baseline Pre-operative scans			
No	127	127	
Yes	135	135	

	Included in BraTS'15	\
We Identified Baseline Pre-operative scans		
No	127	
Yes	135	

	BraTS'15 suggested a baseline pre-op
scan	
We Identified Baseline Pre-operative scans	
No	
38	
Yes	
109	

```
[8]: TCGA_LGG_2015_radiomics_imaging = TCGA_LGG_2015_radiomics.  
      ↪loc[TCGA_LGG_2015_radiomics["We Identified Baseline Pre-operative_  
      ↪scans"]=="Yes"]  
TCGA_LGG_2015_radiomics.groupby(["We Identified Baseline Pre-operative scans"]).  
      ↪count()
```

```
[8]:
```

	Collection	TCGA ID	\
We Identified Baseline Pre-operative scans			
No	91	91	
Yes	108	108	

	Included in BraTS'15	\
We Identified Baseline Pre-operative scans		
No	91	

Yes

108

BraTS'15 suggested a baseline pre-op

scan

We Identified Baseline Pre-operative scans

No

4

Yes

55

```
[9]: print("Number of samples with radiomics and imaging for the LGG (WHO 2015): ",  
      ↪len(TCGA_LGG_2015_radiomics_imaging), " patients")  
print("Number of samples with radiomics and imaging for the GBM (WHO 2015): ",  
      ↪len(TCGA_GBM_2015_radiomics_imaging), " patients")  
print("Total number of samples: ",  
      ↪len(TCGA_LGG_2015_radiomics_imaging)+len(TCGA_GBM_2015_radiomics_imaging), "  
      ↪patients")
```

Number of samples with radiomics and imaging for the LGG (WHO 2015): 108

patients

Number of samples with radiomics and imaging for the GBM (WHO 2015): 135

patients

Total number of samples: 243 patients

```
[10]: ok = True  
for ps in problems_GBM:  
    if ps in TCGA_GBM_2015_radiomics_imaging["TCGA ID"]:  
        print(f"{ps} has no matching entry in the clinical data registry of_  
        ↪GBMs (WHO 2015)")  
        ok = False  
if ok:  
    print("The GBM patients with pre-operative imaging all have matching_  
    ↪entries in the GBM database (WHO 2015)")  
  
ok = True  
for ps in problems_LGG:  
    if ps in TCGA_LGG_2015_radiomics_imaging["TCGA ID"]:  
        print(f"{ps} has no matching entry in the clinical data registry of_  
        ↪LGGs (WHO 2015)")  
        ok = False  
if ok:  
    print("The LGG patients with pre-operative imaging all have matching_  
    ↪entries in the LGG database (WHO 2015)")
```

The GBM patients with pre-operative imaging all have matching entries in the GBM database (WHO 2015)

The LGG patients with pre-operative imaging all have matching entries in the LGG database (WHO 2015)

0.0.3 Mapping the Imaging data to the Clinical entries

Due to the impossibility to download the data from the TCIA website, the NeuroImaging data (e.g., T1, T2, T1c, and FLAIR) together with the paired multilabel lesion segmentations were downloaded from the complete BraTS 2021 dataset. This collection is the most extensive in the literature comprising of patients from other notable collections (e.g., UCSF). Needless to say, we are solely interested in the TCGA-GBM and TCGA-LGG collections.

However, the identification of the imaging data is different than that of the clinical data. Fortunately, there exists a file that allows us to cross-identify all the subjects. Even more, a binary label (0/1) for the MGMT promoter makes a further quality check possible: We will cross examine the MGMT values between databases to make sure no information was altered.

```
[11]: BraTS_To_TCIA = pd.read_csv("BraTS-2021_Mapping-ID_TCIA-Collections.csv",
    ↪sep=',')
BraTS_To_TCGA_GBM = BraTS_To_TCIA.loc[BraTS_To_TCIA["Data Collection (as on
    ↪TCIA+additional)"]=="TCGA-GBM"]
BraTS_To_TCGA_LGG = BraTS_To_TCIA.loc[BraTS_To_TCIA["Data Collection (as on
    ↪TCIA+additional)"]=="TCGA-LGG"]
```

```
[ ]: img_dir = "BraTS-ID_TCGA-GBM_set-"
for patient_data in tqdm(BraTS_To_TCGA_GBM.iloc, total=len(BraTS_To_TCGA_GBM),
    ↪desc="Renaming patients"):
    TCIA_ID = patient_data["PatientID on TCIA Radiology Portal"]
    BraTS2021_ID = patient_data["BraTS2021 ID"]
    set = patient_data["Segmentation (Task 1) Cohort"]
    MGMT_BraTS = patient_data["MGMT value"]

    os.makedirs(f"TCGA-GBM/{TCIA_ID}", exist_ok=True)
    files = glob.glob(f"{img_dir}{set}/{BraTS2021_ID}/*")
    for fl in files:
        shutil.copy(fl, f"TCGA-GBM/{TCIA_ID}")
        modality = fl.split(f"{BraTS2021_ID}")[-1]
        Path(f"TCGA-GBM/{TCIA_ID}/{BraTS2021_ID}{modality}").rename(f"TCGA-GBM/
    ↪{TCIA_ID}/{TCIA_ID}{modality}")
```

```
[12]: BraTS_To_TCGA_GBM.head()
```

```
[12]: Data Collection (as on TCIA+additional) Site ID \
76 TCGA-GBM 5.0
78 TCGA-GBM 5.0
80 TCGA-GBM 8.0
81 TCGA-GBM 6.0
82 TCGA-GBM 10.0
```

```

PatientID on TCIA Radiology Portal Study date (m/d/yyyy) per PatientID \
76 TCGA-02-0085 30/01/1999
78 TCGA-02-0102 15/12/1997
```

80	TCGA-12-1094	15/11/1999
81	TCGA-06-1802	03/04/2007
82	TCGA-19-2631	02/03/2003

	BraTS2021 ID	Segmentation (Task 1) Cohort	MGMT (Task 2) Cohort	\
76	BraTS2021_00100	Training	Training	
78	BraTS2021_00102	Training	Training	
80	BraTS2021_00104	Training	Training	
81	BraTS2021_00105	Training	Training	
82	BraTS2021_00106	Training	Training	

	MGMT value
76	1.0
78	0.0
80	0.0
81	1.0
82	1.0

```
[ ]: img_dir = "BraTS-ID_TCGA-LGG_set-"
for patient_data in tqdm(BraTS_To_TCGA_LGG.iloc, total=len(BraTS_To_TCGA_LGG),
    desc="Renaming patients"):
    TCIA_ID = patient_data["PatientID on TCIA Radiology Portal"]
    BraTS2021_ID = patient_data["BraTS2021 ID"]
    set = patient_data["Segmentation (Task 1) Cohort"]
    MGMT_BraTS = patient_data["MGMT value"]

    os.makedirs(f"TCGA-LGG/{TCIA_ID}", exist_ok=True)
    files = glob.glob(f"{img_dir}/{set}/{BraTS2021_ID}/*")
    for fl in files:
        shutil.copy(fl, f"TCGA-LGG/{TCIA_ID}")
        modality = fl.split(f"{BraTS2021_ID}")[1]
        Path(f"TCGA-LGG/{TCIA_ID}/{BraTS2021_ID}{modality}").rename(f"TCGA-LGG/
            {TCIA_ID}/{TCIA_ID}{modality}")
```

```
[13]: BraTS_To_TCGA_LGG.head()
```

	Data Collection (as on TCIA+additional)	Site ID	\
1148	TCGA-LGG	12.0	
1149	TCGA-LGG	12.0	
1150	TCGA-LGG	12.0	
1151	TCGA-LGG	12.0	
1152	TCGA-LGG	12.0	

	PatientID on TCIA Radiology Portal	Study date (m/d/yyyy)	per PatientID	\
1148	TCGA-CS-4942	22/02/1997		
1149	TCGA-CS-4944	08/02/2001		
1150	TCGA-CS-5393	06/06/1999		

1151	TCGA-CS-5396	02/03/2001
1152	TCGA-CS-5397	15/03/2001

	BraTS2021 ID	Segmentation (Task 1)	Cohort	MGMT (Task 2)	Cohort \
1148	BraTS2021_01476		Training		NaN
1149	BraTS2021_01477		Training		NaN
1150	BraTS2021_01478		Training		NaN
1151	BraTS2021_01479		Training		NaN
1152	BraTS2021_01480		Training		NaN

	MGMT value
1148	NaN
1149	NaN
1150	NaN
1151	NaN
1152	NaN

0.0.4 Identification of glioblastomas according to the 2021 WHO classification

The TCGA identified glioblastomas (GBMs) according to the WHO 2016 criteria, which was largely based on histology rather than molecular and genetic profiling. Therefore, we redefine the clinical data based on the new guidelines. The header of the csv metadata file will also be reduced to maintain the most relevant information to ensure compatibility with the other datasets.

The reference that the TCGA follows is from 2015, however, the criteria are the same as in 2016.

```
[75]: GBM_metadata_with_imaging = TCGA_GBM_2015.loc[TCGA_GBM_2015["patient"]
      ↪isin(TCGA_GBM_2015_radiomics_imaging["TCGA ID"])]
```

```
[76]: TCGA_GBM_clinical_2021 = pd.DataFrame(columns=[
      "patient",
      "Gender",
      "Age..years.at.diagnosis.",
      "Grade",
      "diagnosis.WHO2021",
      "Survival..months.",
      "Vital.status..1.dead.",
      "IDH.status",
      "MGMT.promoter.status",
      "Karnofsky.Performance.Score"
    ])

data_list = [] # Use a list for better performance

for _, entry in GBM_metadata_with_imaging.iterrows():
    new_entry = {column: entry[column] for column in TCGA_GBM_clinical_2021.
      ↪columns if column != "diagnosis.WHO2021"}
```

```

# Assign diagnosis based on conditions
if entry["IDH.code1.subtype"] == "IDHwt" and new_entry["Grade"] == "G4":
    new_entry["diagnosis.WHO2021"] = "Glioblastoma, IDH-wildtype"
elif entry["IDH.code1.subtype"] == "IDHmut-non-code1" and
↪new_entry["Grade"] == "G4":
    new_entry["diagnosis.WHO2021"] = "Astrocytoma, IDH-mutant"
else:
    new_entry["diagnosis.WHO2021"] = np.nan # Use `np.nan` instead of
↪`None` for pandas compatibility

data_list.append(new_entry)

# Convert list to DataFrame in one step
TCGA_GBM_clinical_2021 = pd.DataFrame(data_list)

```

```
[99]: TCGA_GBM_clinical_2021.groupby(["diagnosis.WHO2021"], dropna=False).count()
```

```
[99]:
```

	patient	Gender	Age..years.at.diagnosis.	Grade	\
diagnosis.WHO2021					
Astrocytoma, IDH-mutant	4	4		4	4
Glioblastoma, IDH-wildtype	103	103		103	103
NaN	28	27		27	27

	Survival..months.	Vital.status..1.dead.	\
diagnosis.WHO2021			
Astrocytoma, IDH-mutant	4		4
Glioblastoma, IDH-wildtype	103		103
NaN	27		27

	IDH.status	MGMT.promoter.status	\
diagnosis.WHO2021			
Astrocytoma, IDH-mutant	4		4
Glioblastoma, IDH-wildtype	103		65
NaN	5		7

	Karnofsky.Performance.Score	
diagnosis.WHO2021		
Astrocytoma, IDH-mutant		4
Glioblastoma, IDH-wildtype		85
NaN		25

```
[101]: TCGA_GBM_clinical_2021_confirmed_diagnosis = TCGA_GBM_clinical_2021.
↪loc[(TCGA_GBM_clinical_2021["diagnosis.WHO2021"]=="Astrocytoma, IDH-mutant")
↪| (TCGA_GBM_clinical_2021["diagnosis.WHO2021"]=="Glioblastoma,
↪IDH-wildtype")]

root = "../Glioblastoma_TCGA-GBM_v1-20170717"
```

```
os.makedirs(os.path.join(root, "data"), exist_ok=True)
TCGA_GBM_clinical_2021_confirmed_diagnosis.to_csv(os.path.join(root, "data/
↳TCGA-GBM_clinical_v1.0.csv"), index=None)
```

0.1 Now we move the imaging data with confirmed diagnostic and clinical co- variates

```
[113]: os.makedirs(os.path.join(root, "TCGA-GBM_data-raw_V1"), exist_ok=True)
for id in tqdm(TCGA_GBM_clinical_2021_confirmed_diagnosis["patient"]):
    os.makedirs(f"{root}/TCGA-GBM_data-raw_V1/{id}", exist_ok=True)
    for f in glob.glob(f"./TCGA-GBM/{id}/*"):
        shutil.copy(f, os.path.join(root, f"TCGA-GBM_data-raw_V1/{id}"))
```

```
0%|          | 0/107 [00:00<?, ?it/s]
100%|        | 107/107 [00:17<00:00, 6.21it/s]
```